

Putnam POTD November 17, 2025

I. PROBLEM

(2005 B4) For positive integers m and n , let $f(m, n)$ denote the number of n -tuples $(x_1, x_2, \dots, x_n)$ of integers such that $|x_1| + |x_2| + \dots + |x_n| \leq m$. Show that $f(m, n) = f(n, m)$.

II. SOLUTION

We consider the difference

$$f(m, n) - f(m-1, n) = |\{(x_1, \dots, x_n) \mid |x_1| + \dots + |x_n| = m\}|,$$

Let $D = |\{(x_1, \dots, x_n) \mid |x_1| + \dots + |x_n| = m\}|$, and consider the values x_n can take. If $x_n = 0$, then only need to consider $|\{(x_1, \dots, x_{n-1}) \mid |x_1| + \dots + |x_{n-1}| = m\}|$. If $x_n \neq 0$, then $|x_n| \geq 1$, so we need to consider $2|\{(x_1, \dots, x_{n-1}) \mid |x_1| + \dots + |x_{n-1}| \leq m-1\}|$. Thus

$$D = |\{(x_1, \dots, x_{n-1}) \mid |x_1| + \dots + |x_{n-1}| = m\}| + 2|\{(x_1, \dots, x_{n-1}) \mid |x_1| + \dots + |x_{n-1}| \leq m-1\}|.$$

By definition, note that

$$|\{(x_1, \dots, x_{n-1}) \mid |x_1| + \dots + |x_{n-1}| = m\}| = f(m, n-1) - f(m-1, n-1)$$

and

$$|\{(x_1, \dots, x_{n-1}) \mid |x_1| + \dots + |x_{n-1}| \leq m-1\}| = f(m-1, n-1)$$

so

$$f(m, n) - f(m-1, n) = f(m, n-1) - f(m-1, n-1) + 2f(m-1, n-1).$$

$$f(m, n) = f(m, n-1) + f(m-1, n) + f(m-1, n-1).$$

Now we show from induction that $f(n, m) = f(m, n)$. We first show that $f(n, 1) = f(1, n)$ as the base case. $f(n, 1) = |\{x_1 \mid |x_1| \leq n\}| = 2n + 1$. $f(1, n) = |\{(x_1, \dots, x_n) \mid |x_1| + \dots + |x_n| \leq 1\}|$. For the $= 1$ case, there are $2n$ choices, and for the $= 0$ choice, there is one option, so the result is also $2n + 1$, and thus they are equal.

From the IH assume that $f(n, m) = f(m, n)$ for all $m + n < k$, where $k \geq 3$ is an integer, then consider $m + n = k$, since

$$f(m, n) = f(m, n-1) + f(m-1, n) + f(m-1, n-1)$$

and

$$f(n, m) = f(n, m-1) + f(n-1, m) + f(n-1, m-1) = f(m-1, n) + f(m, n-1) + f(m-1, n-1) = f(m, n)$$

for all $m + n = k$, and we are done with the proof.